Exploring a connection between air quality and labour productivity

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Air quality



Hypothesis:

Air quality and labour productivity are correlated on global scale

Air quality index

First index 1966





Air quality index

- Measuring pollutant concentrations
- SO2, NO2, CO, O3, **PM2.5**, PM10 …
- Many different indexes, **IQAir**, EAQI

PM2.5 legend

0-5	5.1-10	10.1-15	15.1-25	25.1-35	35.1-50	>50.1
Meets WHO guideline	Exceeds by 1 to 2 times	Exceeds by 2 to 3 times	Exceeds by 3 to 5 times	Exceeds by 5 to 7 times	Exceeds by 7 to 10 times	Exceeds by over 10 times

Rank	Country/Region	2024	2023	2022	2021	2020	2019	2018	Population
1	Chad	91.8	--	89.7	75.9	--	--	--	17,179,740
2	Bangladesh	78	79.9	65.8	76.9	77.1	83.3	97.1	169,356,251
3	Pakistan	73.7	73.7	70.9	66.8	59	65.8	74.3	231,402,117

Labour productivity

- Labour productivity – GDP/hour worked
- Labour productivity → Air Quality : public policies, industry, low infrastructure
- Air Quality → Labour productivity : health benefits, less sick days, better cognitive skills

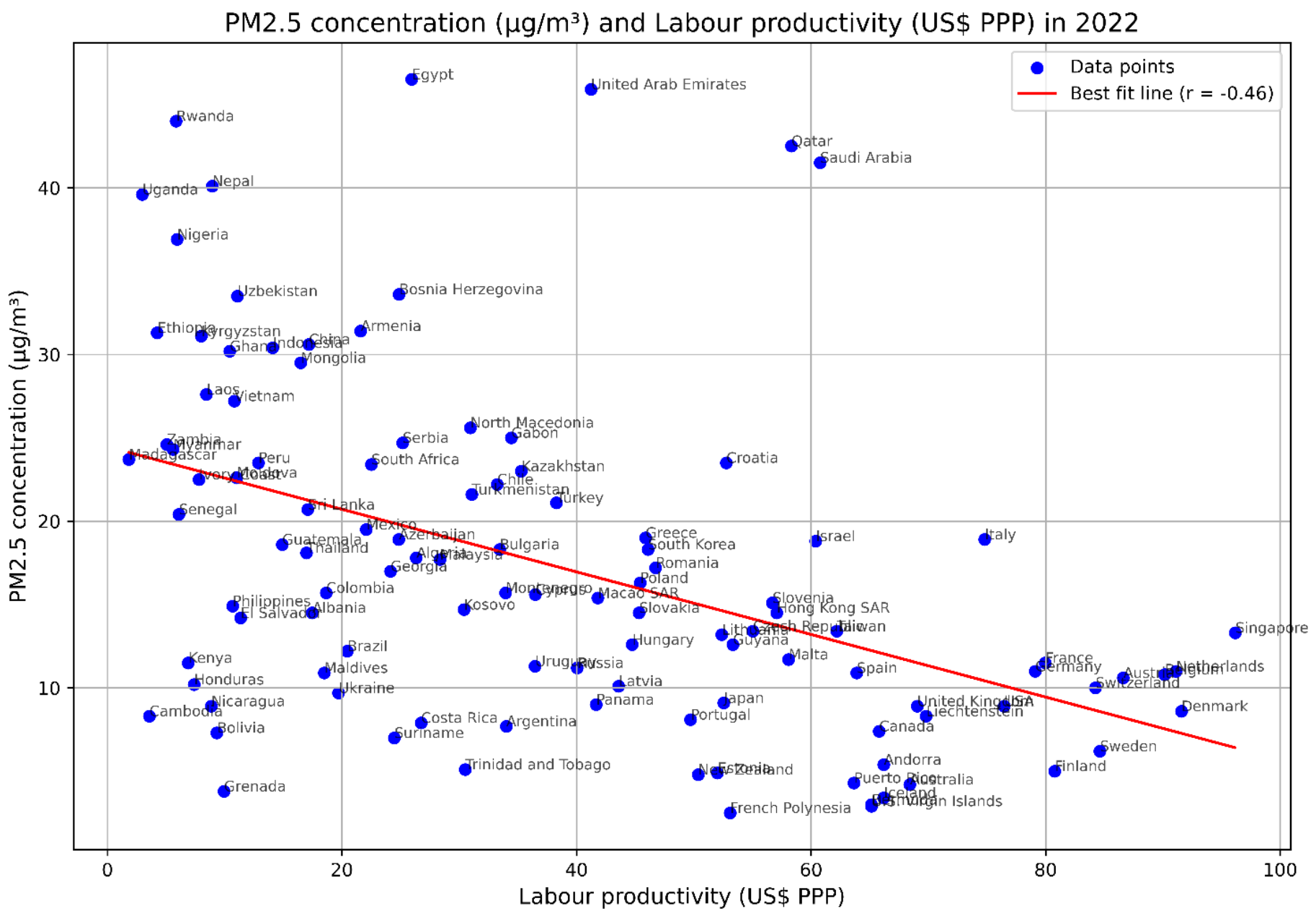
Previous studies

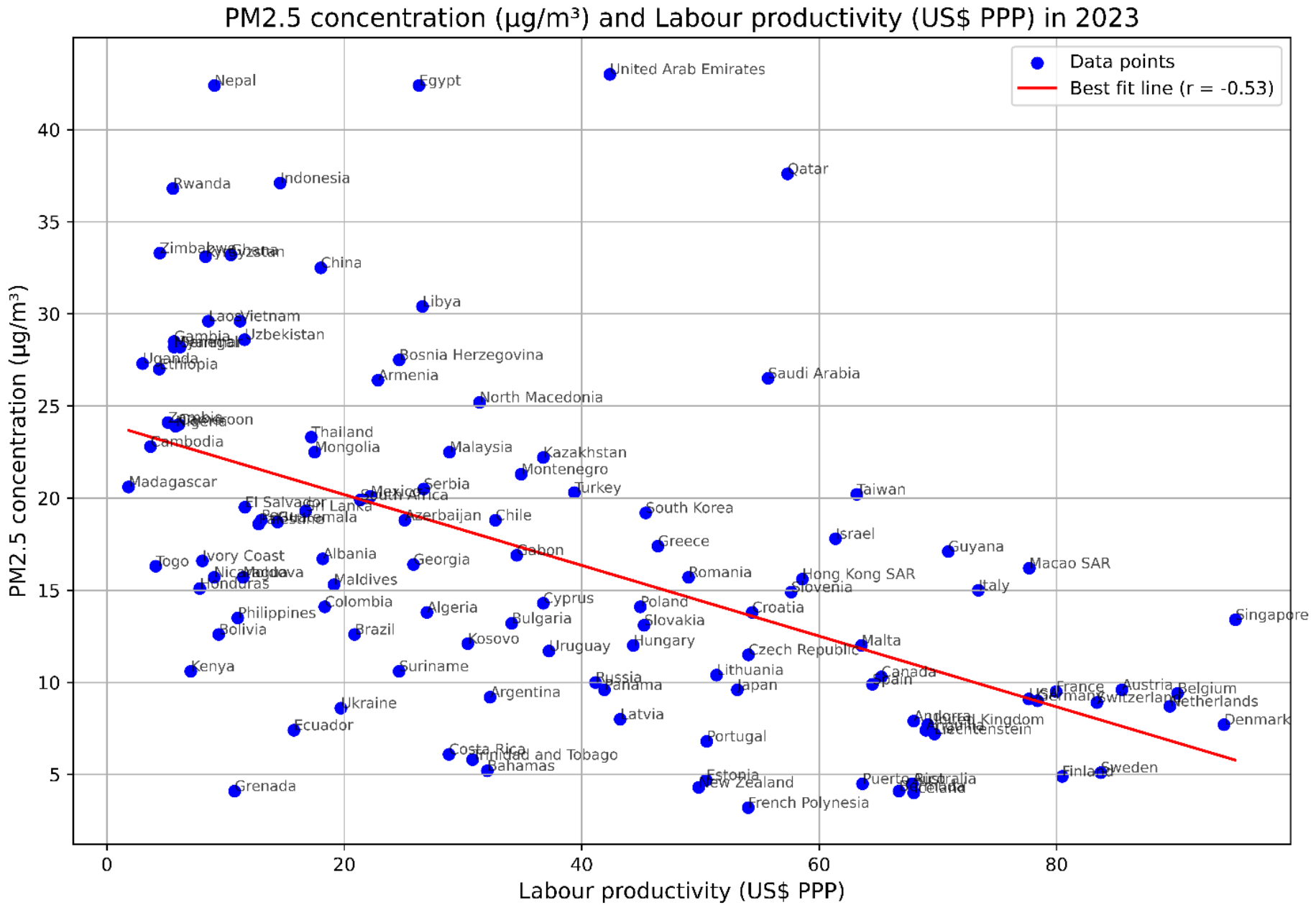
- **EU** - 1µg/m3 increase in PM2.5 causes a 0.8% reduction in real GDP
- **China** - a 10-unit increase in the air pollution index decreases the number of daily calls handled by a worker by 0.35 percent
- **France** – a 10% increase in fine particulate matter leads, on average, to a 1.5% decrease in labour productivity

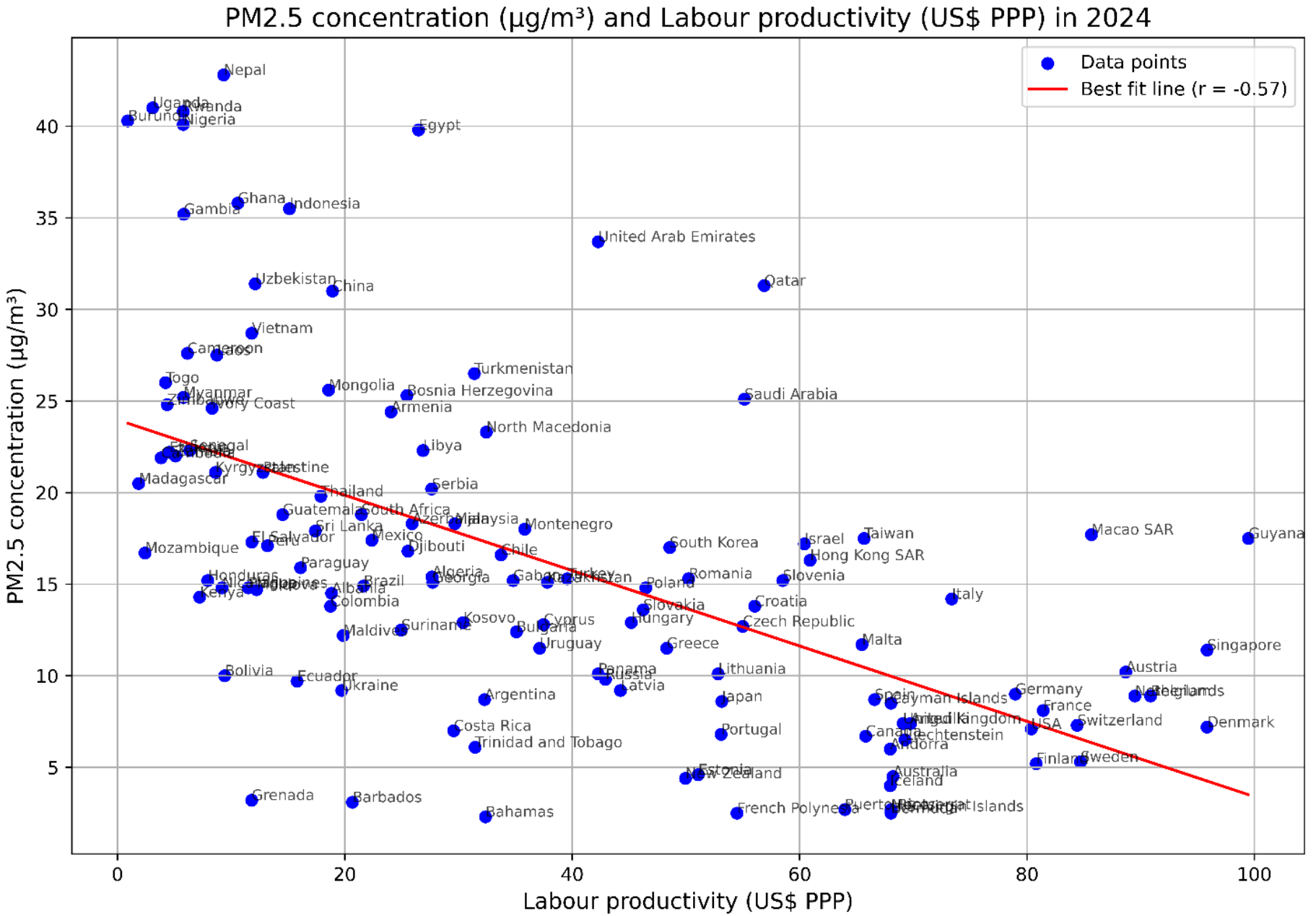
Data

- 138 countries
- 2022,2023,2024

Year	2024	
Country	PM2.5	Labour productivity
Bahamas	2.3	32.36
French Polynesia	2.5	54.48
Bermuda	2.5	68.03
U.S. Virgin Islands	2.6	68.03
Montserrat	2.7	68.03
Puerto Rico	2.7	63.97
Barbados	3.1	20.66
Grenada	3.2	11.8
Iceland	4	67.96
New Zealand	4.4	49.95
Australia	4.5	68.2
Estonia	4.6	51.08
Finland	5.2	80.8
Sweden	5.3	84.7
Andorra	6	67.96
Trinidad and Tobago	6.1	31.44
Liechtenstein	6.5	69.26
Canada	6.7	65.81
Portugal	6.8	53.09
Costa Rica	7	29.57
USA	7.1	80.36
Denmark	7.2	95.83


PM2.5 concentration (µg/m³) and Labour productivity (US$ PPP) in 2022
Data points
Best fit line (r = -0.46)
PM2.5 concentration (µg/m³)
Labour productivity (US$ PPP)
0
20
40
60
80
100
10
20
30
40
Egypt
United Arab Emirates
Rwanda
Qatar
Saudi Arabia
Nepal
Uganda
Nigeria
Uzbekistan
Bosnia Herzegovina
Armenia
Ethiopia
Kyrgyzstan
Ghana
Indonesia
China
Mongolia
Laos
Vietnam
North Macedonia
Gabon
Zambia
Myanmar
Serbia
Madagascar
Peru
South Africa
Croatia
Kazakhstan
Chile
Turkmenistan
Turkey
Senegal
Sri Lanka
Mexico
Azerbaijan
Guatemala
Thailand
Bulgaria
Greece
South Korea
Israel
Italy
Romania
Georgia
Poland
Colombia
Macao SAR
Slovenia
Philippines
El Salvador
Albania
Kosovo
Slovakia
Hong Kong SAR
Singapore
Hungary
Guyana
Malta
Brazil
Kenya
Maldives
Uruguay
Russia
Spain
France
Germany
Netherlands
Honduras
Latvia
Switzerland
Ukraine
Nicaragua
Panama
Japan
Denmark
Cambodia
Costa Rica
Argentina
Portugal
Canada
Bolivia
Suriname
Sweden
Andorra
Trinidad and Tobago
Finland
Grenada
Puerto Rico
Australia
Iceland
French Polynesia


PM2.5 concentration (µg/m³) and Labour productivity (US$ PPP) in 2023
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PM2.5 concentration (µg/m³)
Labour productivity (US$ PPP)
0
20
40
60
80
5
10
15
20
25
30
35
40
Nepal
Egypt
United Arab Emirates
Qatar
Rwanda
Indonesia
China
Libya
Vietnam
Uzbekistan
Bosnia Herzegovina
Armenia
Saudi Arabia
North Macedonia
Thailand
Cambodia
Mongolia
Malaysia
Kazakhstan
Montenegro
Madagascar
Serbia
Turkey
Taiwan
South Africa
El Salvador
Sri Lanka
Azerbaijan
Chile
South Korea
Israel
Greece
Guyana
Togo
Ivory Coast
Albania
Georgia
Gabon
Romania
Hong Kong SAR
Macao SAR
Slovenia
Italy
Honduras
Maldives
Cyprus
Poland
Croatia
Philippines
Colombia
Algeria
Bulgaria
Slovakia
Singapore
Bolivia
Brazil
Kosovo
Uruguay
Hungary
Malta
Czech Republic
Kenya
Suriname
Lithuania
Canada
Spain
Russia
Panama
Japan
France
Austria
Belgium
Switzerland
Netherlands
Argentina
Ukraine
Latvia
Andorra
United Kingdom
Denmark
Ecuador
Portugal
Costa Rica
Trinidad and Tobago
Bahamas
Sweden
Finland
Estonia
New Zealand
Australia
Grenada
French Polynesia


PM2.5 concentration (µg/m³) and Labour productivity (US$ PPP) in 2024
Data points
Best fit line (r = -0.57)
PM2.5 concentration (µg/m³)
Labour productivity (US$ PPP)
0
20
40
60
80
100
5
10
15
20
25
30
35
40
Nepal
Uganda
Rwanda
Burundi
Nigeria
Egypt
Ghana
Gambia
Indonesia
United Arab Emirates
Uzbekistan
China
Qatar
Vietnam
Cameroon
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Turkmenistan
Mongolia
Bosnia Herzegovina
Saudi Arabia
Myanmar
Ivory Coast
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North Macedonia
Libya
Madagascar
Serbia
Thailand
Guatemala
South Africa
Sri Lanka
Montenegro
Mexico
Djibouti
South Korea
Israel
Taiwan
Macao SAR
Guyana
Mozambique
Chile
Hong Kong SAR
Paraguay
Honduras
Algeria
Romania
Slovenia
Brazil
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Ukraine
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Japan
Spain
Germany
Cayman Islands
France
Switzerland
Costa Rica
Portugal
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Results

- P-value under 0.05 suggesting correlation between the concentration of PM2.5 and labour productivity.
- More productive countries tend to have cleaner air.

Year	2022
Correlation coefficient	0.457711479
R Squared	0.209499798
Intercept	24.46884366
Slope	-0.187742976
P value	4.97192E-07
Standard Error	9.30104086

Year	2023
Correlation coefficient	0.531622601
R Squared	0.28262259
Intercept	24.02487391
Slope	-0.191959928
P value	6.94336E-10
Standard Error	7.960423087

Year	2024
Correlation coefficient	0.573003939
R Squared	0.328333514
Intercept	23.96789092
Slope	-0.20571059
P value	2.36012E-12
Standard Error	7.887937008